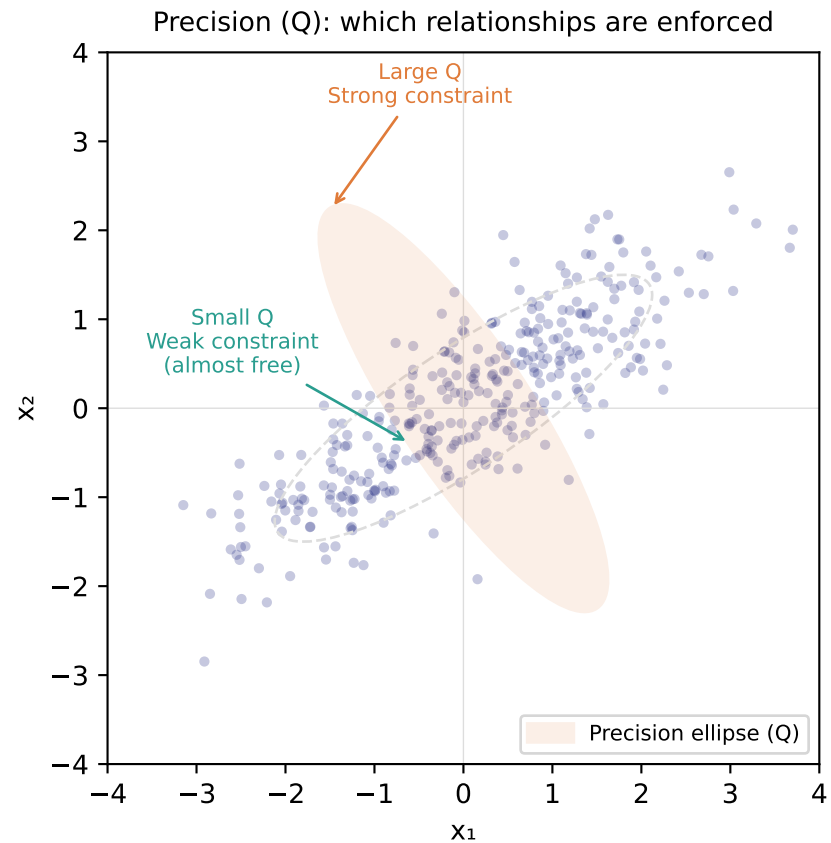
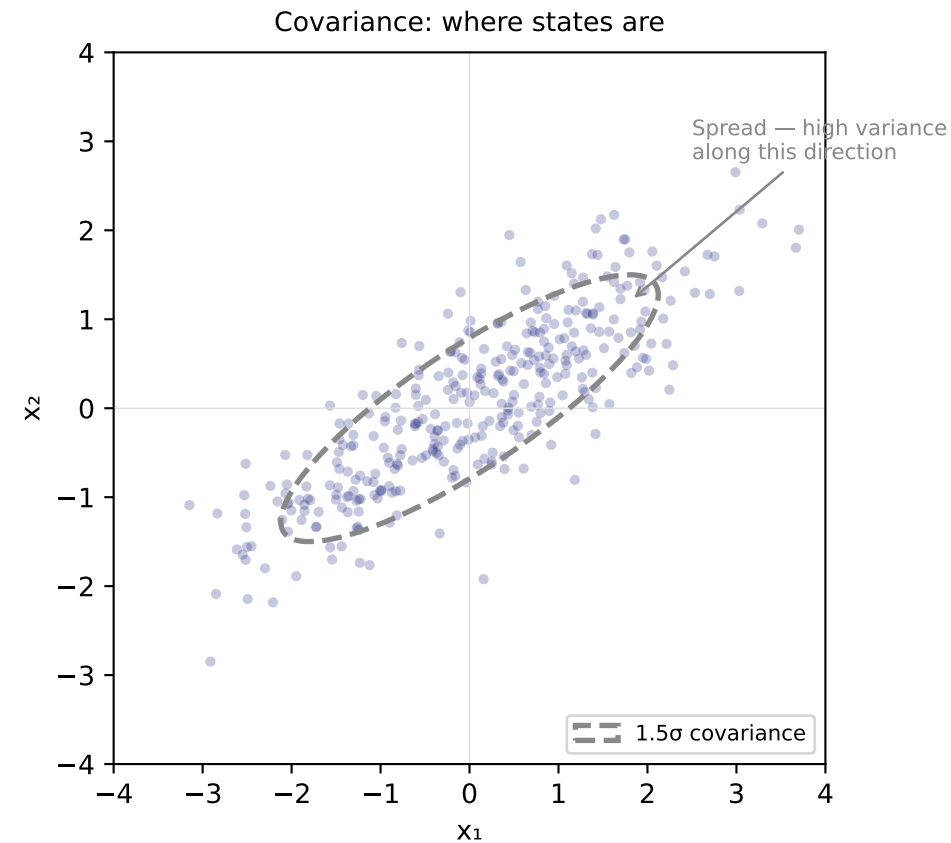
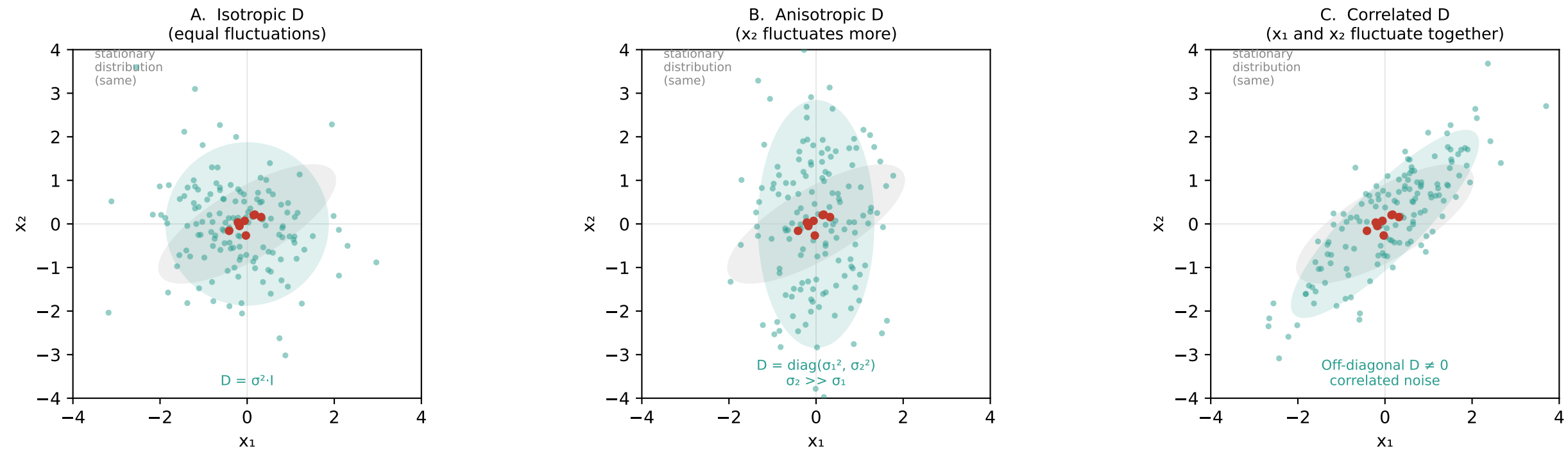


Figure 1 — What Q Sees



*Q is the inverse covariance. It measures which conditional relationships are enforced, not where states lie.
A large off-diagonal $Q[i,j]$ means i and j are directly coupled — even after removing shared influences.*

Figure 2 — What D Sees



*D encodes where fluctuations enter. The stationary distribution (gray, same in all panels) is set by Q .
D changes independently: the same stationary state can arise from isotropic, anisotropic, or correlated noise.*

Worm: D reorganizes between roaming (teal) and dwelling (red) · $\rho(D_{\text{roam}}, D_{\text{dwell}}) \approx 0.14$

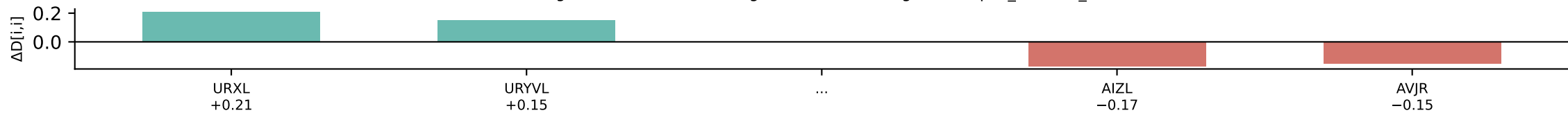
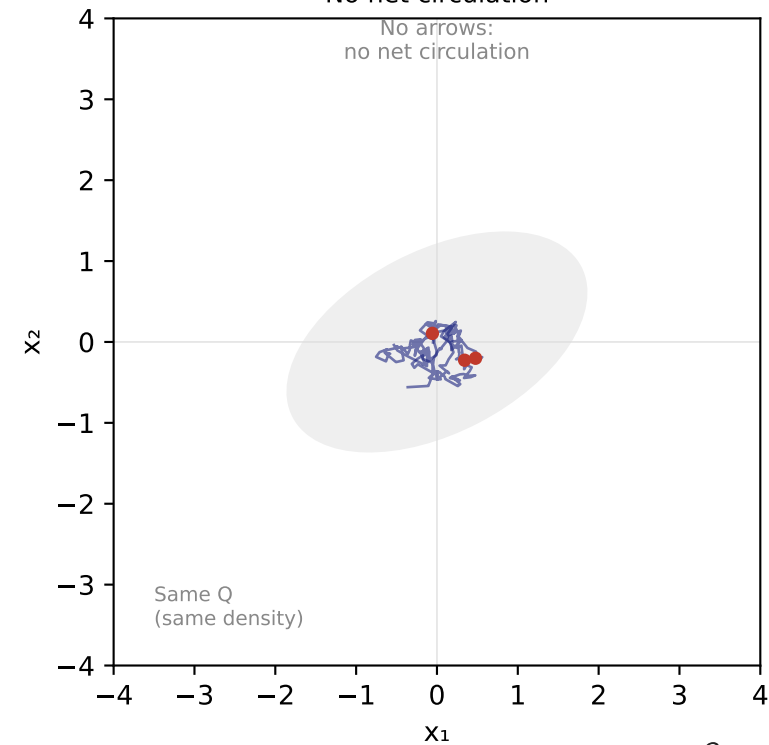
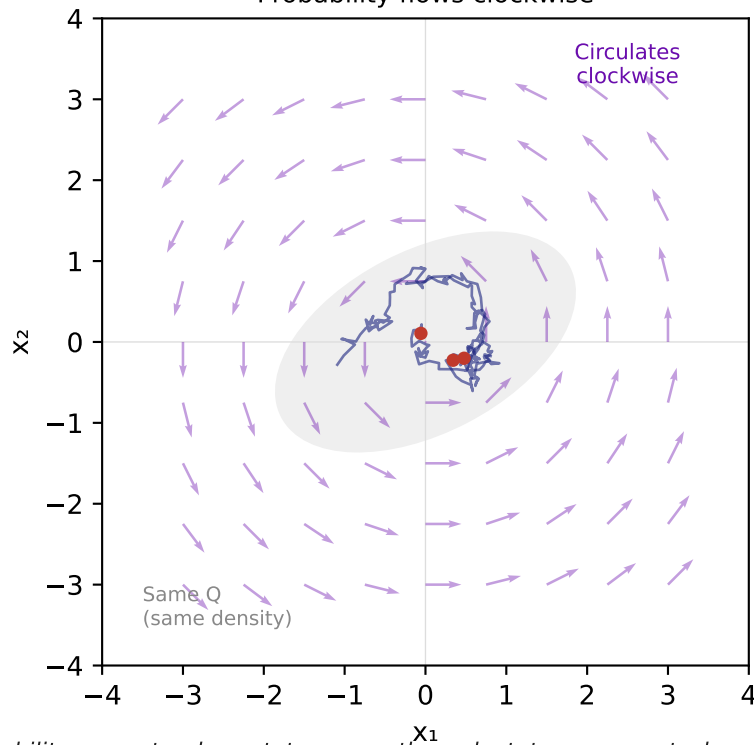


Figure 3 — What Ω Sees

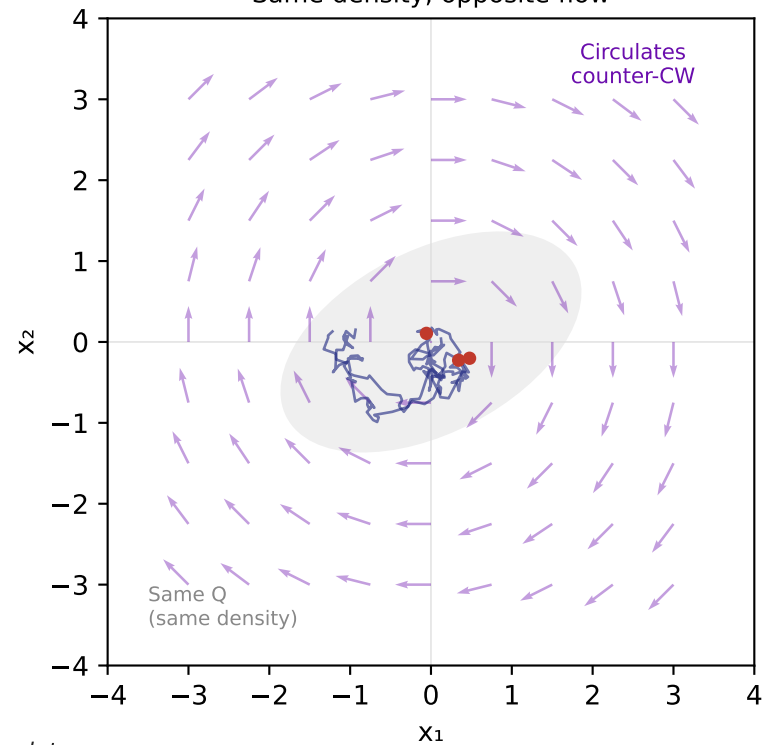
A. Detailed balance ($\Omega = 0$)
No net circulation



B. Circulating current ($\Omega \neq 0$)
Probability flows clockwise



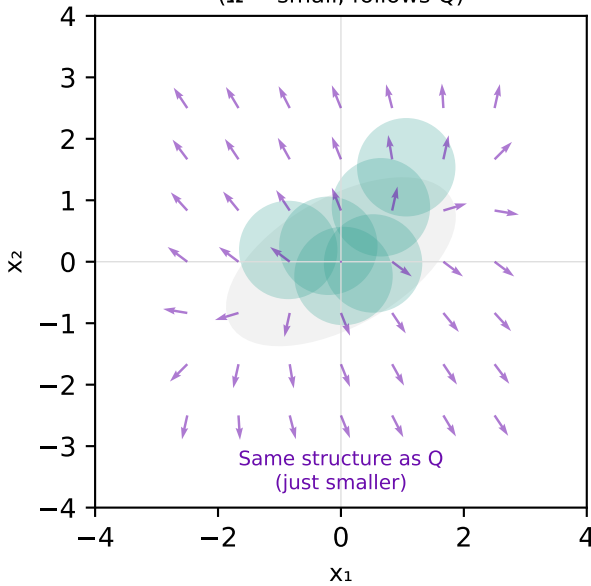
C. Time-reversed ($-\Omega$)
Same density, opposite flow



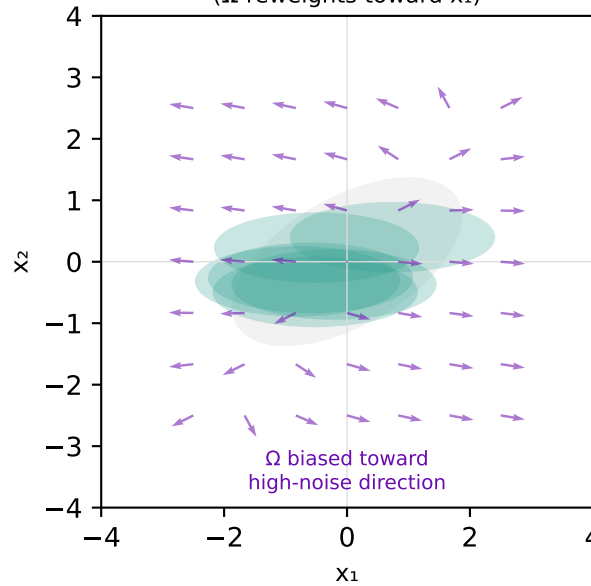
Ω measures probability current — how states move through state space, not where they accumulate.
All three panels have identical stationary distributions (same Q). Only the flow pattern (Ω) differs.

Figure 4 — Combining Fluctuations and Organization: $\Omega = DQ + A$

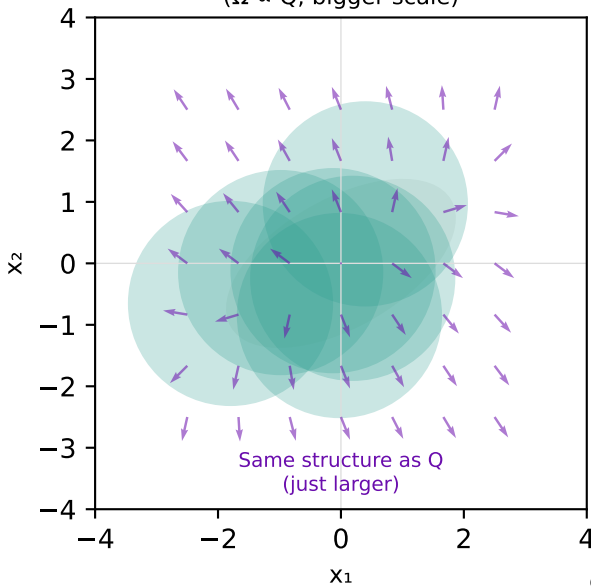
A. Uniform small D
($\Omega \approx$ small, follows Q)



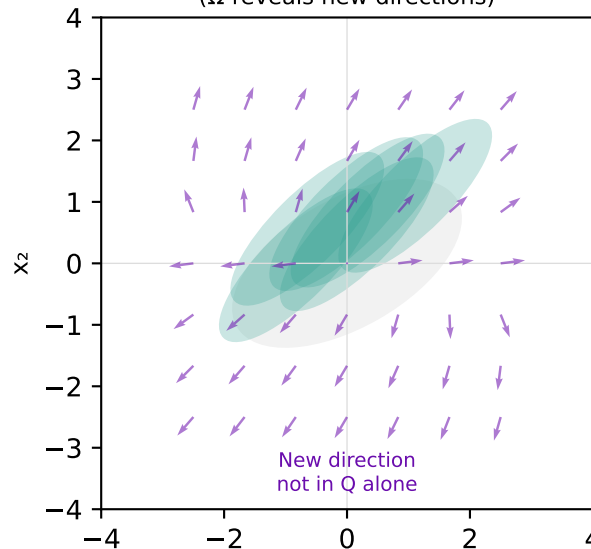
B. Anisotropic D
 x_1 gets more noise
(Ω reweights toward x_1)



C. Uniform large D
($\Omega \propto Q$, bigger scale)



D. Correlated D
(off-diagonal non-zero)
(Ω reveals new directions)



Visual Metaphor

Q

Road map
(where are the roads?)

D

Traffic volume
(how many cars per lane?)

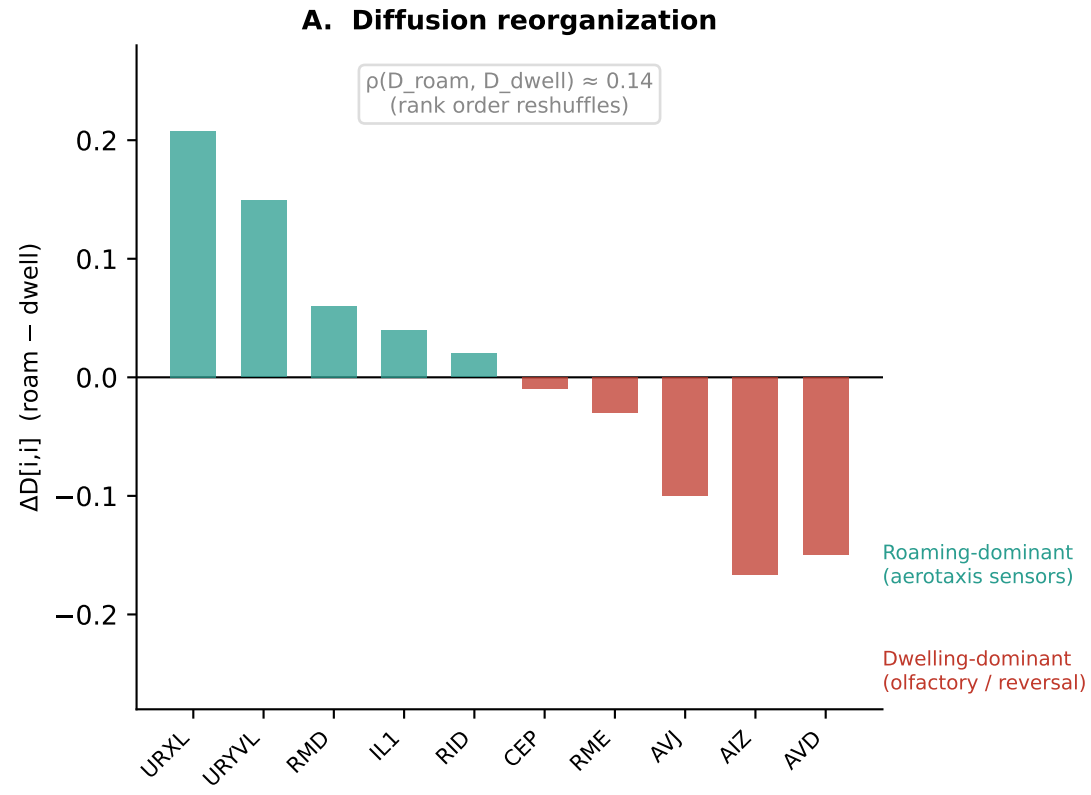
Ω

Traffic organization
(where does traffic flow?)

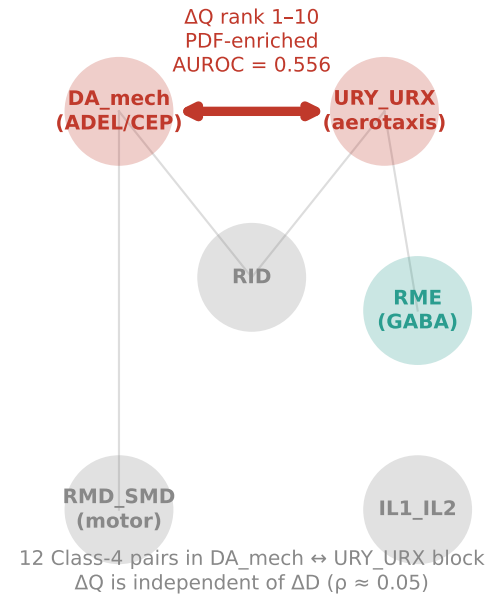
*Same road map (Q).
Different traffic (D).
Different flow (Ω).*

$\Omega = DQ + A$. When D is uniform, Ω is a scaled copy of Q (panels A, C).
When D is non-uniform or correlated, Ω reveals new structure (panels B, D).

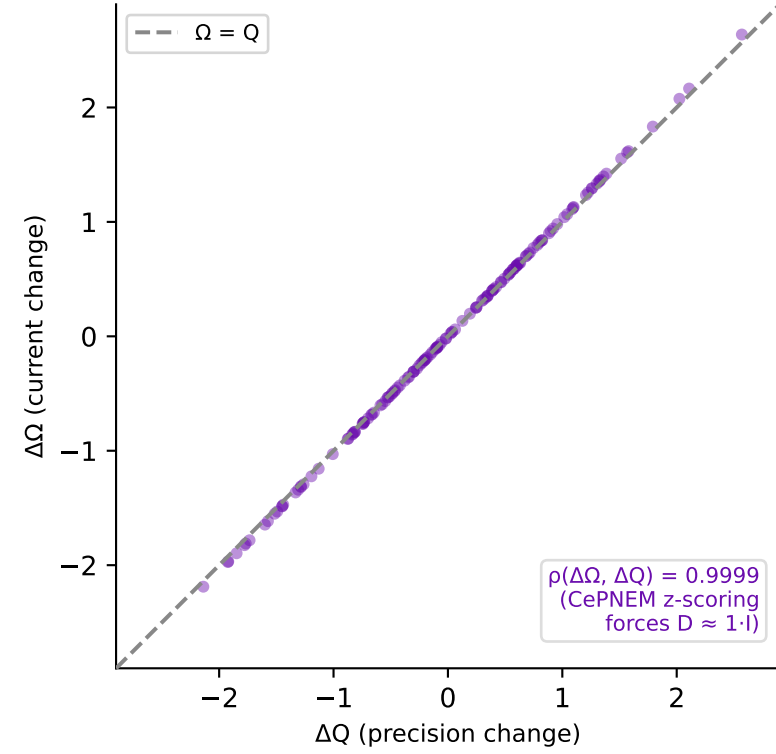
Figure 5 — What Happened in C. elegans?



B. Precision reorganization



**C. Current convergence
(Ω collapses to Q)**



D changes (Panel A). *Q* changes (Panel B). But $\Delta \Omega \approx \Delta Q$ (Panel C).
z-scoring forces *D* $\approx$ constant·*I*, making Ω a near-constant rescaling of *Q*.
The biological interpretation rests on ΔQ — Ω adds nothing here.

Figure 6 — When Ω Matters

